101 SQL

Apr 2023

Download MySQL

Windows: https://dev.mysql.com/downloads/installer/

MacOS (DMG): https://dev.mysql.com/downloads/mysql/

+ https://dev.mysql.com/downloads/workbench/

macOS説明:
https://dev.mysql.com/doc/refman/8.0/en/windows-installation-layout.html

Structured Query Language

- NOT a “programming” language
- Relational database vs non-relational database

mongoDB

Microsoft SQL Server

ORACLE

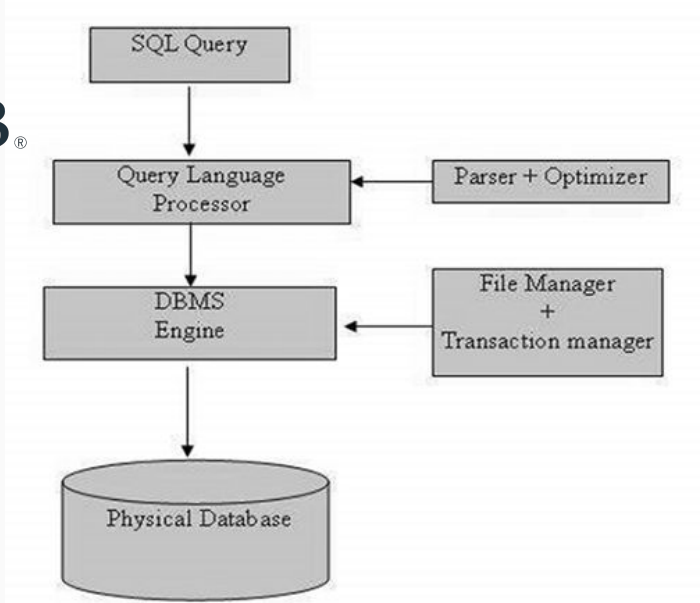
方言: PL/SQL, T-SQL

MariaDB

MySQL

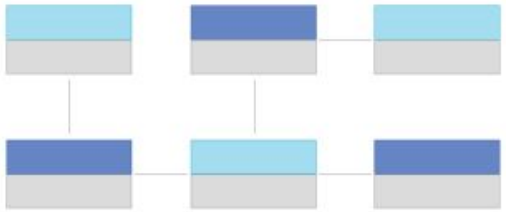
Amazon RDS

PostgreSQL

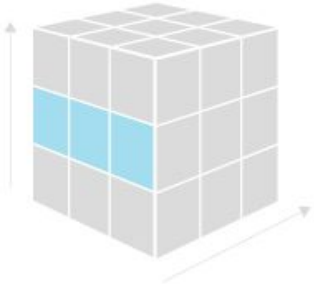


SQL

Relational

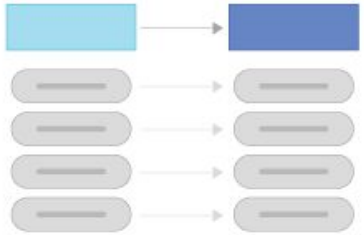


Analytical

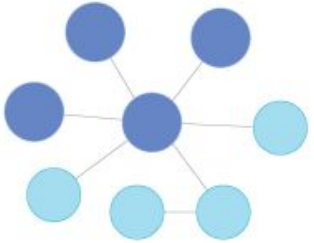


NoSQL

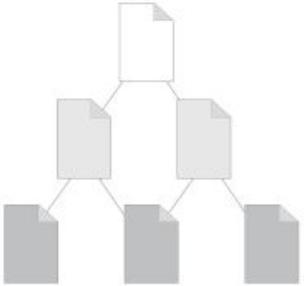
Key - Value



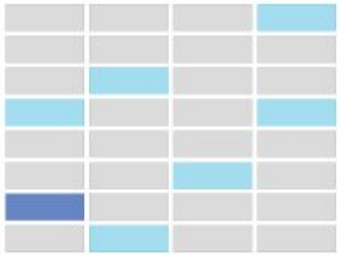
Graph



Document



Wide Column



Create Database

```
CREATE DATABASE testDB;

SHOW DATABASES;
```

Drop database

DROP DATABASE testDB;

SHOW DATABASES;

Use database

/* create another database first */

SELECT database();

```
+------------+
| database() |
+------------+
| NULL       |
+------------+
1 row in set (0.00 sec)
```

USE testDB;

-- select a database

SELECT database();

https://dev.mysql.com/doc/refman/8.0/en/data-types.html
https://www.guru99.com/sql-cheat-sheet.html

Data types (データタイプ)

numeric:

Numeric Data types

Command	Description
`TINYINT( )`	-128 to 127 normal 0 to 255 UNSIGNED.
`SMALLINT( )`	-32768 to 32767 normal 0 to 65535 UNSIGNED.
`MEDIUMINT( )`	-8388608 to 8388607 normal 0 to 16777215 UNSIGNED.
`INT( )`	-2147483648 to 2147483647 normal 0 to 4294967295 UNSIGNED.
`BIGINT( )`	-9223372036854775808 to 9223372036854775807 normal 0 to 18446744073709551615 UNSIGNED.
`FLOAT`	A small approximate number with a floating decimal point.
`DOUBLE( , )`	A large number with a floating decimal point.
`DECIMAL( , )`	A DOUBLE stored as a string , allowing for a fixed decimal point. Choice for storing currency values.

Text Data Types

Command	Description
`CHAR( )`	A fixed section from 0 to 255 characters long.
`VARCHAR( )`	A variable section from 0 to 255 characters long.
`TINYTEXT`	A string with a maximum length of 255 characters.
`TEXT`	A string with a maximum length of 65535 characters.
`BLOB`	A string with a maximum length of 65535 characters.
`MEDIUMTEXT`	A string with a maximum length of 16777215 characters.
`MEDIUMBLOB`	A string with a maximum length of 16777215 characters.
`LONGTEXT`	A string with a maximum length of 4294967295 characters.
`LONGBLOB`	A string with a maximum length of 4294967295 characters.

Date / Time data types

Command	Description
`DATE`	YYYY-MM-DD
`DATETIME`	YYYY-MM-DD HH:MM:SS
`TIMESTAMP`	YYYYMMDDHHMMSS
`TIME`	HH:MM:SS

Create table

```
CREATE TABLE CUSTOMERS(
   ID   INT              NOT NULL,
   NAME VARCHAR (20)     NOT NULL,
   AGE  INT              NOT NULL,
   ADDRESS  VARCHAR (75) ,
   SALARY   DECIMAL (18, 2),
   PRIMARY KEY (ID)
);
```

```
show tables;

desc CUSTOMERS;
```

Drop table

DROP TABLE CUSTOMERS;

戻り値: long

```
MariaDB [testDB]> drop table CUSTOMERS;
Query OK, 0 rows affected (0.011 sec)

MariaDB [testDB]> desc CUSTOMERS;
ERROR 1146 (42S02): Table 'testDB.CUSTOMERS' doesn't exist
MariaDB [testDB]>
```

Insert

```
CREATE TABLE CUSTOMERS(
   ID   INT              NOT NULL    AUTO_INCREMENT,
   NAME VARCHAR (20)     NOT NULL,
   AGE  INT              NOT NULL,
   ADDRESS  VARCHAR (75) ,
   SALARY   DECIMAL (18, 2),
   PRIMARY KEY (ID)
);
```

AUTO_INCREMENT=1

```
INSERT INTO CUSTOMERS
(NAME,AGE,ADDRESS,SALARY)
VALUES ('Cody', 25, 'Delhi', 1500.00);

...

INSERT INTO CUSTOMERS
(NAME,AGE,ADDRESS,SALARY)
VALUES (...),(...),(...);
```

Select

```
SELECT * FROM CUSTOMERS;
```

```
SELECT ID, NAME, AGE FROM CUSTOMERS
WHERE SALARY > 1000;
```

```
SELECT * FROM CUSTOMERS
WHERE AGE < 30 AND SALARY = 2000.00;
```

OR

```
SELECT * FROM CUSTOMERS
WHERE NAME = 'Cody';
```

WHERE **NOT** AGE > 30

Update

UPDATE CUSTOMERS SET SALARY = 5000.00
WHERE NAME = 'Cody';

UPDATE CUSTOMERS

SET SALARY = 10000.00;

```
MariaDB [testDB]> update CUSTOMERS SET SALARY = 5000.00 WHERE ID > 0;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [testDB]>
```

☐ Safe Updates (rejects UPDATEs and DELETEs with no restrictions)

Delete

DELETE FROM CUSTOMERS WHERE ID = 3;

```
MariaDB [testDB]> DELETE FROM CUSTOMERS WHERE ID = 3;
Query OK, 1 row affected (0.002 sec)

MariaDB [testDB]> select * from CUSTOMERS;
+----+----------+-----+---------+----------+
| ID | NAME     | AGE | ADDRESS | SALARY   |
+----+----------+-----+---------+----------+
|  1 | Cody     |  25 | Delhi   | 10000.00 |
|  2 | Khilan   |  25 | Delhi   |  1500.00 |
|  4 | Chaitali |  25 | Mumbai  |  6500.00 |
+----+----------+-----+---------+----------+
3 rows in set (0.000 sec)

MariaDB [testDB]>
```

DELETE FROM CUSTOMERS;

```
MariaDB [testDB]> DELETE FROM CUSTOMERS;
Query OK, 3 rows affected (0.004 sec)

MariaDB [testDB]> select * from CUSTOMERS;
Empty set (0.000 sec)

MariaDB [testDB]>
```